

Aradhya Singh

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SUMMARY

Software Developer with experience building secure full-stack applications, REST APIs, automation tools, and LLM-powered platforms using Python, JavaScript, TypeScript, React, Node.js, SQL, PostgreSQL, Docker, and GitHub Actions. Strong background in authentication workflows, API testing, fraud technology, identity/access controls, and production issue investigation in financial services environments. Comfortable collaborating with technology, QA, security, and operations teams to deliver reliable, tested, production-ready software.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, Java, Bash

Frontend: React.js, Next.js, HTML, CSS

Backend & APIs: Node.js, Express.js, Spring Boot, REST APIs, Prisma, Hibernate

Databases: PostgreSQL, MySQL, MongoDB, Redis, schema design, query optimization

AI & Internal Tools: OpenAI API, LLM workflows, prompt-to-SQL, Python automation

Auth & Security: Auth0, bcrypt, AES-256 encryption, RBAC, rate limiting

DevOps: Docker, GitHub Actions, CI/CD, Linux, AWS, Git

Testing: Postman, JUnit, Selenium, Charles Proxy, Jira, TestRail, API testing, unit testing, functional testing, regression testing

Documentation: REST API documentation, technical documentation, API specifications

EXPERIENCE

Empire Life

Toronto, ON

Software Engineer

Sept 2026 – Present

- Developing and maintaining Customer Identity and Access Management (CIAM) solutions using Auth0, supporting secure authentication, authorization, and user access workflows.

Canadian Imperial Bank of Commerce

Toronto, ON

Business Analyst, Fraud Technology — Python Automation

Jan 2024 – Present

- Built **Python automation scripts and reconciliation pipelines** using pandas, NumPy, SQL, and PostgreSQL to identify SLA breaches, queue mismatches, duplicate fraud cases, and workflow anomalies, reducing manual review effort by approximately **40%**.
- Developed internal tooling to support fraud case assignment, investigator queue monitoring, exception handling, and operational reporting across high-volume fraud workflows.
- Investigated production workflow issues by analyzing transaction patterns, case routing logic, API responses, and data inconsistencies across CNP, tap, chip, debit, and e-Transfer fraud channels.

Fibra Inc.

Toronto, ON

Software Engineer Intern — Backend (Part-Time)

Jan 2023 – Jun 2023

- Designed and built **RESTful backend services** using Node.js, Express.js, and PostgreSQL, including authentication endpoints, user-management APIs, normalized database schemas, and backend validation logic for the product's beta launch.
- Implemented authentication and identity controls using **JWT**, bcrypt hashing, request validation, and role-based access policies, securing all backend routes and standardizing authorization logic.
- Optimized PostgreSQL schema design and query performance through indexing and normalization,

improving average response time by approximately **35%**.

Score Media and Gaming (Penn Entertainment)

Toronto, ON

Quality Engineering Intern

Jan 2023 – Sep 2023

- Designed and executed automated unit, functional, and regression test suites using **Python**, covering REST API responses, payment scenarios, geolocation checks, and regulatory compliance logic across Android, iOS, and web platforms.
- Used **Charles Proxy**, **Postman**, **Jira**, **TestRail**, and CI/CD pipeline logs to debug API requests, reproduce defects, and validate fixes across UAT, staging, and production environments.
- Automated QA reporting workflows with **Python** by generating TestRail test cases and Jira defect tickets from structured failure outputs, reducing manual reporting effort by approximately **50%**.

PROJECTS

NL-to-SQL Engine

OpenAI API, TypeScript, React, Next.js, Node.js, PostgreSQL, Prisma, Redis, Docker, JWT

- Built a full-stack **LLM-powered platform** using the OpenAI API to convert plain-English prompts into executable SQL queries, with a React/Next.js front-end and Node.js/PostgreSQL backend.
- Developed REST API routes for authentication, database connection management, prompt processing, SQL generation, query execution, and structured error handling.
- Implemented secure multi-tenant access controls using **JWT authentication**, AES-256 credential encryption, Redis rate limiting, and row-level access policies.
- Added SQL validation and guardrails to reduce unsafe query execution, improve response reliability, and protect connected database credentials.
- Containerized the application with **Docker** and configured environment-based deployment settings for local development and production hosting.

Transaction Risk API

Java, Spring Boot, Kafka, Redis, PostgreSQL, Hibernate, Docker, GitHub Actions

- Built a production-ready real-time fraud detection backend using **Java and Spring Boot**, exposing REST APIs that score transactions and return approve, flag, or block decisions based on configurable rule sets.
- Implemented an **event-driven architecture using Kafka** to consume and process payment events asynchronously, publishing fraud decision outcomes under simulated high-throughput workloads.
- Containerized the application with Docker and configured **GitHub Actions CI/CD** to automate build, test, and deployment pipelines end-to-end.
- Designed PostgreSQL persistence using Hibernate ORM; added **Redis caching** for fraud rule lookups to reduce database reads and improve scoring latency under load.

EDUCATION

York University

Toronto, ON

Bachelor of Science in Computer Science

Sep 2019 – Apr 2024